

MINSI LU

Tel: (+86) 188-0013-9136 | Mail: minsilu0330@gmail.com | [Website](#) | [Github](#) | [Google Scholar](#)

Addr: 30 Shuangqing Rd, Haidian District, Beijing, China

EDUCATION BACKGROUND

Department of Pharmacy science, **Tsinghua University**
Bachelor of Science in Pharmaceutical Sciences
Bachelor of Science with minor in Software Engineering

Beijing, China
Aug. 2019 - Present
Sep. 2021 - Present

PUBLICATIONS

Bowen Gao, Bo Qiang, Haichuan Tan, Minsi Ren, Yinjun Jia, **Minsi Lu**, Jingjing Liu, Weiying Ma, Yanyan Lan, 'DrugCLIP: Contrastive Protein-Molecule Representation Learning for Virtual Screening', **NeurIPS 2023**.

Tangqi Fang, Yifeng Liu, Addie Woicik, **Minsi Lu**, Anupama Jha, Xiao Wang, Gang Li, Borislav Hristov, Zixuan Liu, Hanwen Xu, William S. Noble, Sheng Wang, 'Enhancing Hi-C Contact Matrices for Loop Detection with Capricorn', **Bioinformatics 2024**.

RESEARCH EXPERIENCE

Research Assistant, School of Computer Science & Engineering, University of Washington **Seattle, US**
Project: Resolution Enhancement and Time Dimension Modeling for Hi-C Data Jun. 2023 - Oct. 2023

Advisor: Prof. William Stafford Noble, Prof. Sheng Wang

- Aimed to integrate the time and other feature dimensions of Hi-C data for resolution enhancement.
- Integrated the temporal dimension of Hi-C data for better performance.
- Developed Capricorn, a tool for Hi-C resolution enhancement that incorporates chromatin features as additional views of the input Hi-C contact matrix and leverages the conditional diffusion probability model Imagen as a backbone to generate a high-resolution matrix. Capricorn outperforms the state of the art in a cross-cell-line setting.
- Formed a related paper to introducing **Capricorn** for Hi-C resolution enhancement, published in *Bioinformatics*: https://academic.oup.com/bioinformatics/article/40/Supplement_1/i471/7700867

Research Assistant, Institute for AI Industry Research, Tsinghua University **Beijing, China**
Project: Contrastive Learning for Drug Virtual Screening Feb. 2023 - Jun. 2023

Advisor: Prof. YanYan Lan

- Aimed to develop a new method for virtual screening, which is a critical step in computer-aided drug discovery (CADD).
- Developed and evaluated the novel contrastive learning framework, DrugCLIP, which reformulating virtual screening as a dense retrieval task and aligning representations of binding protein pockets and molecules from a large quantity of pairwise data without explicit binding-affinity scores.
- Formed a related paper introducing the **DrugCLIP** for drug virtual screening and feature representation, accepted by *NeurIPS*: https://proceedings.neurips.cc/paper_files/paper/2023/file/8bd31288ad8e9a31d519fdeede7ee47d-Paper-Conference.pdf

Research Assistant, Integrated Science Research Center, Peking University **Beijing, China**
Project: High-throughput Genome-wide Functional Screening Based on CRISPR Jul. 2021 - Sep. 2021

Advisor: Prof. Wensheng Wei

- Aimed to use CRISPR-Cas9 functional screening tools to discover the genes related to the IFN- α -induced apoptosis pathway.
- Designed sgRNA sequences, and construct sgRNA plasmids for gene screening.
- Constructed GoldenGate point mutation on plasmids.
- Carried out cell culture, virus extraction, and titer determination.

Research Assistant, School of Pharmaceutical Science, Tsinghua University **Beijing, China**
Project: Research on Her2 and CXCR4 Bispecific Antibodies Oct. 2020 - Feb. 2021

Advisor: Prof. Juanjuan Du

- Aimed to design a novel bispecific antibody to treat breast cancer.
- Screened the strain, did the transformation, and plasmid transfection, determined the antibody Kd value.

PROJECTS

Tsinghua University

Beijing, China

Project: Multiomics Integration Via Graph Learning

Sep. 2022 - Jan. 2023

Advisor: Prof. Jianyang Zeng

- Aimed to integrate scRNA-seq data and scATAC-seq data through graph learning based on GLUE.
- Decayed the weight of edges according to the distance of the ATAC peak from the TSS and used a set of aggregators to combine the messages in each GCN layer.
- Our method achieved comparative performance with GLUE on metrics, and had the ability to identify more genes with regulatory relationships than GLUE.
- Repo: <https://github.com/minsilu/Multiomics-Integration-via-Graph-Learning>

Project: PPI Network Guided Driver Target Discovery

Sep. 2022 - Jan. 2023

Advisor: Prof. Jianyang Zeng

- Aimed to design a new method for driver target discovery using single-cell RNA-seq data by introducing a protein-protein interaction (PPI) network.
- Introduced PPI to re-calculate regularization loss in the sc-ETM model and design a 2-layer MLP for target prediction.
- Achieved better cell clustering performance than sc-ETM and better driver target discovery performance thanks to the introduction of a PPI network for gene embedding. Biological case studies on our discovered targets also showed the effectiveness of our proposed framework.
- Repo: <https://github.com/minsilu/PPI-Guided-Driver-Target-Discovery>

Project: Text Sentimental Classification Based on the Twitter Dataset

Feb. 2023 - Jun. 2023

Advisor: Prof. Mingsheng Long

- Spearheaded a comprehensive text classification project utilizing a Twitter sentiment analysis dataset, focusing on advanced preprocessing and diverse model comparison.
- Basically used TF-IDF methods for feature extraction; implemented and optimized a variety of machine learning models, including Decision Trees, Random Forests, MLPs, ResNet, and BERT, for nuanced sentiment analysis.
- Achieved high accuracy, particularly with the BERT model, outperforming other models in classifying sentiments into four categories.
- Repo: <https://github.com/minsilu/Text-Sentimental-Classification-based-on-Twitter-Dataset>

Project: Gomoku Genius: AI Search Strategies for Classic Board Game Mastery

Feb. 2023 - Jun. 2023

Advisor: Prof. Mingsheng Long

- Aimed to investigate and implement five distinct AI agents for board games, addressing the challenges of search complexity and function design.
- Optimized **Minimax Search** for Tic-Tac-Toe. Applied **Alpha-Beta Pruning** to reduce search space. Implemented **Truncated Search** in Go-Moku, balancing time constraints with strategic depth using pattern evaluation. Conducted gameplay analysis comparing naive **Monte Carlo Tree Search (MCTS)** and Alpha-Beta Pruning. **Enhanced MCTS with evaluation function**, significantly improving board state assessments and decision-making.
- A practical application of advanced AI search strategies, demonstrating superior performance in board games.
- Repo: <https://github.com/minsilu/Gomoku-Genius--AI-Search-Strategies-for-Classic-Board-Game-Mastery>

Project: I/O Scheduling in xv6 Operating System

Sep. 2021 - Jan. 2022

Advisor: Prof. Lijie Wen

- Aimed to enhance the I/O scheduling capabilities in xv6-2020, adapting to its transition from IDE to virtio disks and integrating advanced I/O scheduling algorithms.
- Implemented various I/O scheduling algorithms, including NOOP, CFQ, SSTF, and Deadline.
- Successfully improved xv6 OS's I/O request management, enhancing efficiency and fairness in process handling, and navigated challenges in system-level programming under kernel mode constraints.
- Repo: <https://github.com/minsilu/IO-scheduling>

SKILLS

Mathematics & Theoretical Foundations: Calculus, Linear Algebra, Probability and Statistics, Discrete Mathematics, Data Structure and Algorithms.

Machine Learning & AI Expertise: Knowledge of Machine Learning and Deep Learning principles. Proficient with major Deep Learning frameworks (TensorFlow, PyTorch)

Programming Skills: Skilled at C/C++, Java, Python, R, SQL. Familiar with Linux/Unix.

Domain Expertise in Life Sciences: Background in pharmacy, chemistry, and biology. Hands-on experience with experimental protocols in biomedical research

AWARDS & HONORS

Scholarship for Excellent in Academic, University Scholarship (10%)	2023
First Prize in 16 th "Novozymes Cup" Life Science Culture Festival, Municipal Award (10%)	2021
Third Prize in 39 th "Challenges Cup" Student Academic Competition, University Award	2021
Scholarship for Progress in Academic, University Scholarship	2021
Third Prize in Winter Vacation Social Practice for Students, University Award	2021
Gold Medal in 27 th China Biology Olympiad(CBO), National Award	2018

SOCIETY ROLES

Leader at Student Winter Vacation Social Practice Team.

Dec. 2020 - Jan. 2021